Part 4 Cloud native production

It’s been an incredible journey so far. Chapter after chapter, we have gone through patterns, principles, and best practices for working with cloud native applications, and we’ve built a bookshop system using Spring Boot and Kubernetes. It’s time to prepare for production. Part 4 guides you through the last few steps to make your cloud native applications production-ready, addressing concerns like observability, configuration management, secrets management, and deployment strategies. It also covers serverless and native images.

Chapter 13 describes how to make your cloud native applications observable using Spring Boot Actuator, OpenTelemetry, and the Grafana observability stack. You’ll learn how to configure Spring Boot applications to produce relevant telemetry data, such as logs, health, metrics, traces, and more. Chapter 14 covers advanced configuration and secrets management strategies, including Kubernetes-native options like ConfigMaps, Secrets, and Kustomize. Chapter 15 guides you through the final steps of your cloud native journey and teaches you how to configure Spring Boot for production. You’ll then set up continuous deployment for your applications and deploy them to a Kubernetes cluster in the public cloud, adopting a GitOps strategy. Finally, chapter 16 covers serverless architectures and functions with Spring Native and Spring Cloud Function. You’ll also learn about Knative and its powerful features that provide a superior developer experience on top of Kubernetes.